Transcriptome-wide identification and characterization of NRT/NPF gene family and their response to abiotic stress in *Tectona grandis*

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Introduction

Nitrate (NO_3^-) is the preferred nitrogen source of higher plants that plays a crucial role in determining plant growth and productivity. Nitrate transporters (NRTs) participate in nitrate uptake, assimilation, transport and allocation processes that are within the plant. Plants have evolved two NO_3^- absorption systems to manage the heterogeneity of NO_3^- in the soil, namely a low-affinity transport system (LATS), and a high-affinity transport system (HATS), which are mediated by nitrate transporter 1 (NRT1) and nitrate transporter 2 (NRT2), respectively. NRT1/PEPTIDE TRANSPORTER (PTR) family (NPF) genes, initially characterized as nitrate or peptide transporters in plants, are involved in the transport of a large variety of substrates, including amino acids, nitrate, auxin (IAA), jasmonates (JAs), abscisic acid (ABA) and gibberellins (GAs) and glucosinolates.

Objective

A comprehensive analysis NO3- transporter (NRT/NPF) gene family of *T. grandis* is presented in terms of phylogeny, cis-elements and motif analysis. The expression pattern of differentially expressed genes (DEGs) of NRT was analysed from seven transcriptomic dataset of *T. grandis* that grown under varying climatic conditions such as semi moist areas (1201-1600 mm in rainfall, TNAG02, TSNL23, T51 and TATR2) and very moist areas (2500 mm< in rainfall, TTE09, TDA3 and TDE7).

Results

A total of 73 NRT1/PTR (as known as NPF) and 5 NRT2 genes have been identified from the transcriptome dataset of Teak assembled against the chromosomes in accordance with the whole genome of this species (https://datadryad.org/stash/dataset/doi:10.5061/dryad.77b2422). Non-redundant peptide sequences of NRT1/PTR and NRT2 gene family were retrieved from , *Populus trichocarpa* v3.0 for comparative phylogenetic analysis. NRT1/PTR and NRT2 genes have been identified in *P. trichocarpa* and *T. grandis* are classified into eight subfamilies. NPF orthologues from in *P. trichocarpa* were located together with *T. grandis* NPF members in the same clades representing the eight subfamilies (NPF1–NPF8). Among the NPF gene family, the NPF5 subfamily contains the most (52 members) NPF genes, followed by subfamilies NPF4 (20 members), NPF2 (17 members), NPF8 and NPF1 (14 and 13 members for each subfamily), NPF6 (12 members), NPF2 (11 members), NPF7 (7 members) and NPF3 (6 members) with the least NPF genes.

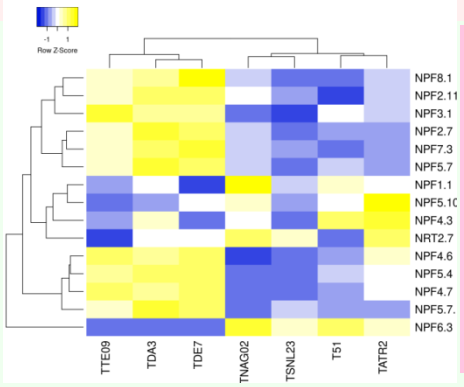
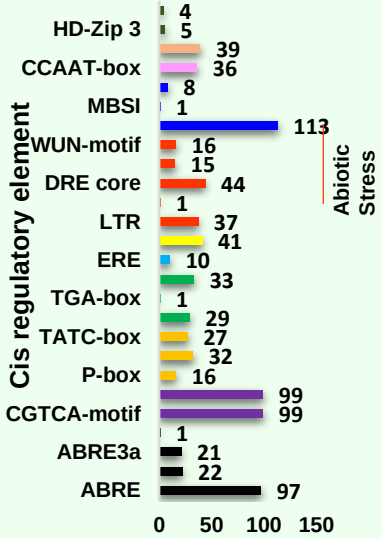


Fig 3. Hierarchical clustering of insilico expression profiles of differentially expressed TgNPF/NRT genes from diverse geographical location of *T. grandis*. Logarthmic of odds fold change is set to 2 and FDR< 0.05 was used as the threshold to identify significant differences in expression between different conditions. The 16 genes were clustered into 4 groups. The samples from different locations (TTE09, TDA3, TDE7, TNAG02, TSNL23, T51 and TATR2 were clustered according to the amount of rainfall (mm) in particular location. The color scale on the left-hand side represents transcript per million (TPM). Gene expression values for each gene were transformed using Z-scores. Higher expression levels are represented in blue, and lower expression levels are denoted in yellow.

Fig 1. Exon–intron structure represented in kilobase (kb) scale of 78 TgNPF/NRT genes and Schematic representation of the **conserved motifs** in the NPF/NRT proteins from Teak.

Fig 2. Cis-acting regulatory elements in the promoter regions of candidate TgNPF/NRT genes.



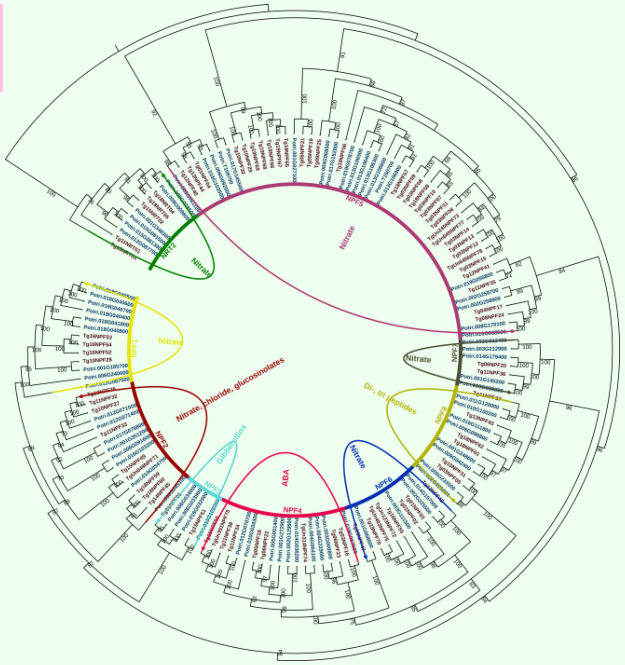


Fig 4. Maximum-likelihood phylogenetic tree rooted of *T. grandis* NRT family proteins were clustered into two major clades: NPF with seven subclades and NRT2. *T. grandis* NPF/NRT sequences were retrieved from the transcriptome data and then manually curated to select unique sequences. Multiple alignment of the peptide sequences between *Populus* and *T. grandis* from the resulting 152 sequences was performed using the MAFFT, and the phylogenetic tree was constructed using IQ-Tree v1.6.9 and ultrafast bootstrap, with 1000 replicates. The resulting 9 NRT families are indicated by colored sections and family name acronyms. Members of *T. grandis* (78) and *P. trichocarpa* (74) NPF/NRT2 protein family were denoted with purple and blue letters, respectively. Bootstrap values are described in the branch points.

Conclusions

In this study, 78 TgNPF/NRTs members were identified that divided into 9 subfamilies having conserved gene and protein structure within each subfamily or clade. In summary, results reveal that expression of TgNPF/NRT may be regulated by various kinds of hormones and abiotic stresses. NPF4 subfamily is mainly involved in abscisic acid transportation, which need to be further characterized.